



# CSI142 Object-Oriented Programming

## Consolidation and Bridge to Methods

**Week 2 • Lecture 6**

Integrate: switch + do-while + ArrayList + args

**Fri 6 Feb 2026 • 08:00–09:00 • Room 252-008**



- I can write a menu with do-while
- I can select options cleanly with switch
- I can store unknown-size input in ArrayList
- I can avoid crashes: bounds checks, args.length checks



- 1) Which loop guarantees at least one run?**
- 2) What happens if break is missing in switch?**
- 3) Last valid index in an array of length n is... ?**
- 4) Which is correct: list.length or list.size() ?**
- 5) What must you check before using args[0] ?**



- **Fall-through:** add `break` (or comment intentional fall-through)
- **Infinite loop:** ensure the loop variable changes, and exit condition is reachable
- **Out-of-bounds:** use `i < length`, never `i <= length`
- **Empty list:** guard division by zero with `isEmpty()`
- **args misuse:** check `args.length` first



- **Menu: 1) Add score 2) Print summary 3) Exit**
- **Store scores in `ArrayList<Integer>`**
- **Teacher mode via args enables “remove last score” (optional)**
- **Validate menu choice, never crash on bad input**



```
1 ArrayList<Integer> scores = new ArrayList<>();
2 boolean teacher = args.length > 0 &&
args[0].equalsIgnoreCase("teacher");
3
4 int choice;
5 do {
6     printMenu(teacher);
7     choice = in.nextInt();
8     switch (choice) {
9         case 1: addScore(scores, in); break;
10        case 2: printSummary(scores); break;
11        case 3: System.out.println("Bye"); break;
12        case 9: if (teacher) removeLast(scores); break;
13        default: System.out.println("Invalid");
14    }
15 } while (choice != 3);
```

## Key points

- Flow first, details later
- Methods reduce repetition (Week 3)
- Teacher mode is optional



- **static void printMenu(boolean teacher)**
- **static void addScore(ArrayList<Integer> scores, Scanner in)**
- **static double average(ArrayList<Integer> scores)**
- **static void printSummary(ArrayList<Integer> scores)**



- **Next week: methods, parameter passing, overloading**
- **Bring your Lab 2 program, we will refactor it into methods**
- **Exit ticket: write one method signature you think your program needs**